

07.1 Figure 3 shows *Gammarus pulex*, commonly known as a freshwater shrimp.

Figure 3

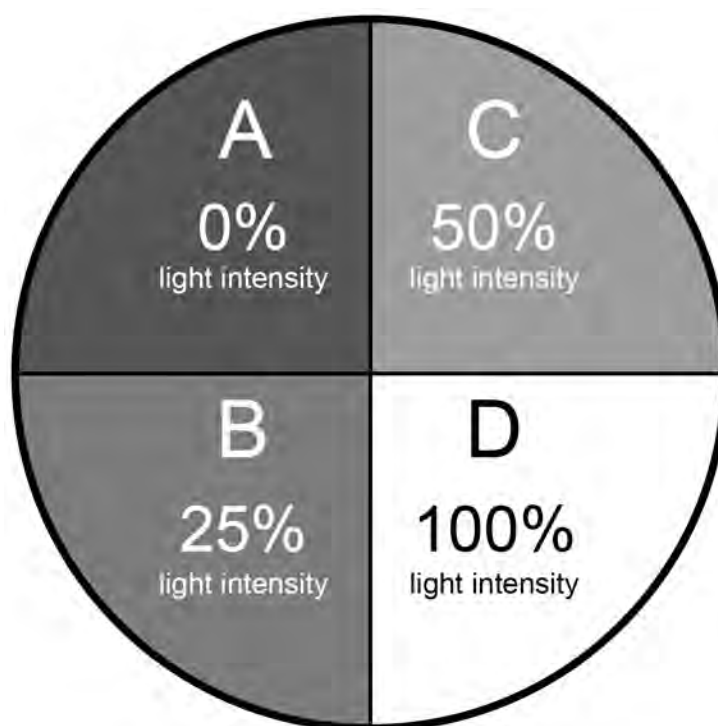


A student used a choice chamber to investigate the effect of light intensity on the movement of *G. pulex*.

The student:

- collected 30 shrimps from a freshwater stream
- added freshwater and gravel substrate from the same stream to each segment
- used layers of black paper on the lid of the choice chamber to reduce the percentage light intensity in 3 segments of the choice chamber as shown in **Figure 4**
- placed 30 shrimps at the centre of the choice chamber
- waited 10 minutes before counting the number of shrimps in each segment.

Figure 4



The student stated that 'the procedure used in this investigation helps to ensure the validity of any conclusions made'.

Suggest and explain **three** features of this investigation that justify this statement.

[3 marks]

1 _____

2 _____

3 _____

Question 7 continues on the next page

Turn over ►

07.2

Table 3 shows the student's results.

Table 3

Segment of choice chamber	Percentage light intensity	Number of <i>G. pulex</i> counted after 10 minutes
A	0	15
B	25	8
C	50	4
D	100	3

The student observed directional movement by *G. pulex*.

Name the type of behaviour shown by *G. pulex* **and** suggest an advantage to *G. pulex* of this behaviour.

Use the information in **Table 3** in your answer.

[2 marks]

Type of behaviour _____

Advantage _____

0 7 . 3 The student analysed the results in **Table 3** using the chi-squared test (χ^2) and calculated a χ^2 value of 11.87

Table 4 shows the critical values of χ^2 at different probability (**P**) values for different degrees of freedom.

Table 4

Degrees of freedom	Critical values of χ^2 at different P values				
	P = 0.95	P = 0.5	P = 0.05	P = 0.01	P = 0.005
1	0.00	0.46	3.84	6.64	7.88
2	0.10	1.39	5.99	9.21	10.60
3	0.35	2.37	7.82	11.35	12.84
4	0.71	3.36	9.49	13.28	14.86
5	1.15	4.35	11.07	15.09	16.75

State a null hypothesis for this investigation **and** explain why the χ^2 test was the appropriate statistical test to use.

[2 marks]

Null hypothesis _____

Explanation for using a χ^2 test _____

Question 7 continues on the next page

[4 marks]

